A note on the soundness of an optimized gemini variant

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Abstract

We give a formal analysis of the optimized variant of the gemini polynomial commitment scheme [BCHO22] used by the Aztec Network [azt]. Our work is motivated by an attack on a previous implementation [GL25].

1 Introduction

[BCHO22] presented a pairing-based polynomial commitment scheme called gemini for multilinear polynomials (ml-PCS). It can in fact be viewed as a reduction from constructing an ml-PCS to a univariate one; which can then be instantiated very efficiently in the pairing setting where we have the univariate scheme of [KZG10].

There are a couple of technicalities, however, that need to be resolved for secure use of gemini as a component of a PLONK style proving system based on multilinear polynomials.

- [BCHO22] assumes the structured reference string (SRS) for [KZG10] is of the exact size of the multilinears being committed. However the SRS we have in practice might be larger. We would ideally want to prove gemini secure with such a larger SRS without needing to resort to additional degree checks.
- PLONK requires an additional opening of the "shifted polynomial" whose values on the boolean hypercube are shifted by one location compared to the actual committed witness polynomial. Lemma 3.9 of Hyperplonk[CBBZ23] provides a method of simulating the shifted polynomial using two evaluations of the original. However, it would be nice to have a PCS that directly supports shift evaluations.

Aztec uses a variant of gemini resolving these two issues and containing an additional optimization that reduces proof length: In the original [BCHO22] construction, the prover sends commitments to univariate polynomials $g_1, \ldots, g_d$, and must send evaluations of each one at two points $r, -r$. By instead using a "tower" of evaluation points

r^{2^i}, one of each two evaluations can be derived by the verifier and does not need to be sent. This optimization was first suggested by Hamelink [ham]. However the original implementation of this optimization was found to be unsound [GL25].

The purpose of this note is to analyze the exact variant of gemini used by Aztec (including a correction following [GL25]) and prove it knowledge sound.

2 Preliminaries

2.1 Basic notation

We denote by $\mathscr{ML}_d$ the space of multi-linear polynomials over $\mathbb{F}$ in d variables. Fix d and let $n := 2^d$. Let $D = 2^k$ be a power of two with $D \geq n$ (n corresponds to the size of polynomials the prover works with, and D to the size of KZG SRS we have at hand.). We denote by $\mathscr{B}_s$ the set of binary strings of length s.

We make the general convention, both for variables and elements of $\mathbb{F}$, that $\bar{z} := 1-z$.

3 Multilinear polynomials

In this section, we present required results and definitions about multilinear polynomials.

We define the well-known **eq** multilinear polynomial in $2d$ variables.

$$\mathbf{eq}(x, y) := \prod_{0 \leq j < d} (x_j y_j + \bar{x}_j \bar{y}_j)$$

We have for $x, y \in \mathscr{B}_d$, $\mathbf{eq}(x, y) = 1$ when $x = y$ and $\mathbf{eq}(x, y) = 0$ otherwise. We use the convention that an integer $0 \leq i < n$ can be used as an input to **eq** by interpreting i as its binary representation. Namely, for $0 \leq i < n$, $u \in \mathbb{F}^d$, $\mathbf{eq}(i, u) := \mathbf{eq}(i_0, \ldots, i_{d-1}, u)$ where $i = \sum_{j=0}^{d-1} i_j 2^j$.

We can use **eq** to define the "multilinear Lagrange polynomials". Namely, for $0 \leq i < n$ let

$$L_i(X_0, \ldots, X_{d-1}) := \mathbf{eq}(i, X_0, \ldots, X_{d-1}) = \prod_{0 \leq j < d} (i_j X_j + \bar{i}_j \bar{X}_j)$$

Note that L_i is the multi-linear polynomial that is one on $(i_0, \ldots, i_{d-1})$ and zero otherwise.

Given a vector $f = (a_0, \ldots, a_{n-1}) \in \mathbb{F}^n$, we define $\mathfrak{ml}_d(f) : \mathbb{F}^d \to \mathbb{F}$ to be the multilinear function on d variables evaluating to f on the Boolean cube $\{0,1\}^d$. That is $\mathfrak{ml}_d(f)(X_0, \ldots, X_{d-1}) := \sum_{i=0}^{n-1} a_i L_i(X_0, \ldots, X_{d-1})$.

Shifted multilinear polynomials Fix any integer $\ell > 0$ and let $L := 2^\ell - 1$. Suppose that $F(X_0, \ldots, X_{\ell-1}) = \mathfrak{ml}_\ell(f)$ where $f = (a_0, \ldots, a_L) \in \mathbb{F}^{L+1}$. We define $F^{\leftarrow}$ to be the multilinear polynomial where the coefficients of F are left-shifted by one and padded with zero, i.e.

$$F^{\leftarrow}(X_0, \ldots, X_{\ell-1}) := \mathfrak{ml}_\ell((a_1, \ldots, a_L, 0)).$$

Polynomials on extended domains Our soundness proof will alternate between d-variate and k-variate multilinears. We denote for $0 \le i < 2^k$, $M_i(X_0, \ldots, X_{k-1})$ as the *k-variate i'th Lagrange*. That is,

$$M_i(X_0, \ldots, X_{k-1}) := \prod_{0 \le j < k} (i_j X_j + \bar{i}_j \bar{X}_j).$$

I.e., the k-variate multilinear that is one on the binary decomposition $i_0, \ldots, i_{k-1}$ of i and zero elsewhere on $\mathscr{B}_k$.

For $f = (a_0, \ldots, a_{D-1}) \in \mathbb{F}^D$ we denote by $\mathfrak{ml}_k(f)$ the k-variate multilinear obtaining f's values on $\mathscr{B}_k$. Namely,

$$\mathfrak{ml}_k(f)(X_0, \ldots, X_{k-1}) := \sum_{i=0}^{D-1} a_i M_i(X_0, \ldots, X_{k-1}).$$

The following two claims relate multilinears on a smaller and larger domain in the non-shifted and shifted case. They will be useful later in our knowledge soundness proof.

Claim 3.1. *Fix any $a \in \mathbb{F}^D$. Let $F(X_0, \ldots, X_{d-1}) := \mathfrak{ml}_d((a_0, \ldots, a_{n-1}))$ and let $G(X_0, \ldots, X_{k-1}) := \mathfrak{ml}_k(a)$. Fix any $u \in \mathbb{F}^d$. Then $F(u) = G(u, 0^{k-d})$*

Proof. It's easy to check that for any $i < n$, $L_i(u) = M_i(u, 0^{k-d})$. Also, for $n \le i < D$ we have $M_i(u, 0^{k-d}) = 0$: For such i there will be $j \ge d$ with $i_j = 1$ and we'll get that the corresponding factor $i_j X_j + \bar{i}_j \bar{X}_j$ in M_i vanishes when substituting $X_j = 0$.

Using this we have

$$F(u) = \sum_{i=0}^{n-1} a_i L_i(u) = \sum_{i=0}^{n-1} a_i M_i(u, 0^{k-d})$$

$$= \sum_{j=0}^{D-1} a_j M_j(u, 0^{k-d}) = G(u, 0^{k-d}).$$

□

Claim 3.2. *Fix any $a \in \mathbb{F}^D$. Suppose that $F^{\leftarrow}(X_0, \ldots, X_{d-1}) = \mathfrak{ml}_d((a_1, \ldots, a_{n-1}, 0))$ and $G^{\leftarrow}(X_0, \ldots, X_{k-1}) = \mathfrak{ml}_k((a_1, \ldots, a_{D-1}, 0))$. Fix any $u \in \mathbb{F}^d$. Then, we have*

$$F^{\leftarrow}(u) = G^{\leftarrow}(u, 0^{k-d}) - a_n \prod_{0 \le i < d} u_i.$$

Proof. As described in proof of Claim 3.1, for any $i < n$, we have $L_i(u) = M_i(u, 0^{k-d})$, and for $n \le i < D$ we have $M_i(u, 0^{k-d}) = 0$.

Using this, we have

$$\begin{aligned}
F^{\leftarrow}(u) &= \sum_{i=1}^{n-1} a_i L_{i-1}(u) = \sum_{i=1}^{n-1} a_i M_{i-1}(u, 0^{k-d}) \\
&= \left(\sum_{j=1}^{D-1} a_j M_{j-1}(u, 0^{k-d}) \right) - a_n M_{n-1}(u, 0^{k-d}) \\
&= \left(\sum_{j=1}^{D-1} a_j M_{j-1}(u, 0^{k-d}) \right) - a_n \prod_{0 \le i < d} u_i \\
&= G^{\leftarrow}(u, 0^{k-d}) - a_n \prod_{0 \le i < d} u_i.
\end{aligned}$$

where $M_{n-1}(u, 0^{k-d}) = \prod_{0 \le i < d} u_i$ since $n-1$ has binary form $1^d 0^{k-d}$.

□

3.1 ml-PCS

We give a definition of a multi-linear polynomial commitment scheme that supports shift evaluations, and only supports opening at a random point (the random point element manifests in the knowledge soundness requirement). Specifically, we define a PCS with shift evaluations that given a commitment to F, enables a prover to convince a verifier that $v = F(u)$ as well as that $v^s = F^{\leftarrow}(u)$ for $v, v^s \in \mathbb{F}, u \in \mathbb{F}^d$.

Definition 3.3. *Consider a fixed integer d and $n = 2^d$. A d-polynomial commitment scheme (with shift evaluations) consists of*

- $\mathsf{gen}(D)$ - *a randomized algorithm that outputs an SRS* srs *of size $D \ge n$.*
- $\mathsf{com}(f, \mathsf{srs})$ - *given a polynomial $F \in \mathscr{ML}_d$ with coefficients $f = (a_0, \ldots, a_{n-1}) \in \mathbb{F}^n$, returns a commitment* cm *to F.*
- *A public coin protocol* $\mathsf{open}(\mathsf{cm}, u, v, v^s)$ *between parties* $\mathbf{P}$ *and* $\mathbf{V}$*, where* $\mathbf{P}$ *knows $F \in \mathscr{ML}_d$ consistent with* cm. $\mathbf{P}$ *and* $\mathbf{V}$ *are both given* cm *(the claimed commitment to F), $u \in \mathbb{F}^d$, $v \in \mathbb{F}$ (the claimed correct opening $F(u)$) and $v^s \in \mathbb{F}$ (the claimed correct opening $F^{\leftarrow}(u)$). At the end of the protocol* $\mathbf{V}$ *outputs* acc *or* rej

such that

- ***Completeness:*** *Fix $u \in \mathbb{F}^d$, $f = (0, a_1, \ldots, a_{n-1}) \in \mathbb{F}^n$ and let $F = \mathtt{ml}_d(f)$. Suppose that* $\mathsf{cm} = \mathsf{com}(f, \mathsf{srs})$. *Then, if* open *is run correctly with values* $\mathsf{cm}, u, v = F(u), v^s = F^{\leftarrow}(u)$, *then* $\mathbf{V}$ *outputs* acc *with probability* $1 - \mathsf{negl}(\lambda)$.
- ***Knowledge soundness in the algebraic group model:*** *There exists an efficient extractor E such that for any algebraic adversary $\mathcal{A}$, the probability of $\mathcal{A}$ winning the following game is* $\mathsf{negl}(\lambda)$ *over the randomness of $\mathcal{A}$,* gen *and $u \in \mathbb{F}^d$.*

1. *Given* srs, $\mathcal{A}$ *outputs cm.*
2. *E, given access to the messages of* $\mathcal{A}$ *during the previous step, outputs* $F \in \mathscr{ML}_d$.
3. *Sample a random* $u \in \mathbb{F}^d$.
4. $\mathcal{A}$ *outputs* $v, v^s \in \mathbb{F}$.
5. $\mathcal{A}$ *takes the part of* $\mathbf{P}$ *in the protocol* open *with inputs* cm, u, v, v^s.
6. $\mathcal{A}$ *wins if*
   - $\mathbf{V}$ *outputs* acc *at the end of the protocol, and*
   - *either* $v \neq F(u)$ *or* $v^s \neq F^{\leftarrow}(u)$.

4 Folded polynomials

As in [BCHO22], we crucially rely on the connection between "folding" univariate polynomials and variable substitution in multilinears. We introduce notation for this purpose and explain this connection in the current section.

Notation for polynomial operations: We associate vectors with univariate polynomials in the following natural way: given $f = (a_0, \ldots, a_{n-1}) \in \mathbb{F}^n$, we denote $f(X) = \sum_{i=0}^{n-1} a_i X^i$. For a univariate polynomial f, define $f^{\leftarrow}(X) := f(X)/X$. We extend this association to *Laurent* polynomials with a $1/X$ term: We associate the vector $g = (c_0, \ldots, c_{D-1}, c_{-1})$ with the Laurent polynomial $g(X) = \sum_{i=-1}^{D-1} c_i X^i$.

Polynomial fold: For a polynomial $f(X) = \sum_{i=0}^{n-1} a_i X^i$, we define $f_{even}(X)$ and $f_{odd}(X)$ as the polynomials defined by the even and odd coefficients of f, respectively. Thus, we have:

$$f_{even}(X^2) = \frac{f(X) + f(-X)}{2}, f_{odd}(X^2) = \frac{f(X) - f(-X)}{2X}$$

Given $x \in \mathbb{F}$, we define

$$\mathtt{fold}(f, x)(X) := \bar{x} f_{even}(X) + x f_{odd}(X).$$

where we remind that $\bar{x} = 1 - x$. Note that folding is well-defined also for Laurent polynomials $f(X) = \sum_{i=-j_1}^{j_2} a_i X^i$ as well.

Given $r, x \in \mathbb{F}$ and polynomial f, let $\alpha = f(r), \bar{\alpha} = f(-r)$ and $\beta = \mathtt{fold}(f, x)(r^2)$. Then,

$$\beta = \bar{x} \frac{\alpha + \bar{\alpha}}{2} + x \frac{\alpha - \bar{\alpha}}{2r}$$

This equality is used in the reverse direction in the protocol to derive $\alpha = f(r)$ from $\bar{\alpha} = f(-r)$ and $\beta = \mathtt{fold}(f, x)(r^2)$, as follows. Assume $\bar{x} r + x \neq 0$, then

$$\alpha = \mathtt{derive}(x, \bar{\alpha}, \beta) := \frac{2r\beta - (\bar{x} r - x)\bar{\alpha}}{\bar{x} r + x}$$

Univariate polynomial fold and its multilinear counterpart: Suppose that $f = (a_0, \ldots, a_{n-1})$ and $F = \mathfrak{ml}_d(f)$. Then $\texttt{fold}(f, x)(X)$ and $F_x(X_1, \ldots, X_{d-1}) := F(x, X_1, \ldots, X_{d-1})$ will "have the same coefficients". More precisely,

$$\mathfrak{ml}_{d-1}(\texttt{fold}(f, x)) = F_x$$

In general, for any $\ell \leq d$ and $x = (x_0, \ldots, x_{\ell-1})$, define

$$\texttt{fold}(f, x) := \texttt{fold}(\cdots \texttt{fold}(f, x_0) \cdots, x_{\ell-1})$$

and

$$F_x(X_\ell, \ldots, X_{d-1}) := F(x_0, \ldots, x_{\ell-1}, X_\ell \ldots, X_{d-1}).$$

Then,

$$\mathfrak{ml}_{d-\ell}(\texttt{fold}(f, x)) = F_x.$$

Consider an integer ℓ and $L = 2^\ell - 1$. We define $\mathfrak{ml}_\ell(\cdot)$ for a Laurent polynomial $g = \sum_{i=-1}^{L} c_i X^i$ as $\mathfrak{ml}_\ell(g) := \mathfrak{ml}_\ell((c_0, \ldots, c_L))$. Note that we discard in particular the $1/X$ coefficient when defining the corresponding multilinear.

Given $f = (a_0, \ldots, a_{D-1}) \in \mathbb{F}^D$ and $f^{\leftarrow}(X) := f(X)/X$, let $f_\ell^s = \texttt{fold}(f_0^s, x)$. Denote $f_\ell^s(X) = \sum_{i=-1}^{D'} c_i X^i$ where $D' := 2^{k-\ell} - 1$. Let $G^{\leftarrow} := \mathfrak{ml}_k((a_1, \ldots, a_{D-1}, 0))$. Then, the above definition of $\mathfrak{ml}_\ell(\cdot)$ for Laurent polynomials gives

$$G^{\leftarrow}(x_0, \ldots, x_{\ell-1}, X_\ell, \ldots, X_{k-1}) = \mathfrak{ml}_{k-\ell}(f_\ell^s) = \mathfrak{ml}_{k-\ell}((c_0, \ldots, c_{D'})).$$

5 The main scheme

Univariate PCS: We'll assume as a component a univariate PCS as in the Definition 2.3 in [BDFG20]. We use in implementation the construction in Section 4 of that paper, that we refer to here as SHPLONK.

We are now ready to specify our PCS.

5.1 The Aztec gemini variant

The ml-PCS is the tuple $(\mathsf{gen}, \mathsf{com}, \mathsf{open})$ where

- $\mathsf{gen}(D)$ - Choose uniform $x \in \mathbb{F}$. Output $\mathsf{srs} = ([1]_1, [x]_1, \ldots, [x^{D-1}]_1, [1]_2, [x]_2)$.
- $\mathsf{com}(f, \mathsf{srs})$ - For a d-variate multilinear F with coefficients $f = (a_0, \ldots, a_{n-1}) \in \mathbb{F}^n$, $\mathsf{com}(f, \mathsf{srs}) := [f(x)]_1$.
- $\underline{\mathsf{open}(\mathsf{cm}, u \in \mathbb{F}^d, v \in \mathbb{F}, v^s \in \mathbb{F})}$: An interactive protocol between **P** and **V** with common inputs cm, u, v, v^s where v, v^s are the claimed values, and **P**'s private input is $f \in \mathbb{F}^n$ (with $F = \mathfrak{ml}_d(f)$) such that $\mathsf{cm} = \mathsf{com}(f, \mathsf{srs})$.

1. $\mathbf{V}$ samples and sends $\rho \in \mathbb{F}$ to $\mathbf{P}$.
2. $\mathbf{P}$ computes $f^{\leftarrow}(X) = f(X)/X$ and sets $g_0 = f + \rho \cdot f^{\leftarrow}$.
3. $\mathbf{P}$ computes the polynomials $g_i(X) := \texttt{fold}(g_{i-1}, u_{i-1})$ for $i \in [d]$.
4. For each $i \in [d-1]$, $\mathbf{P}$ sends to $\mathbf{V}$ $\mathsf{cm}_i := \mathsf{com}(g_i, \mathsf{srs})$.
5. $\mathbf{V}$ chooses and sends to $\mathbf{P}$ a random $r \in \mathbb{F}$.
6. $\mathbf{P}$ sends to $\mathbf{V}$ $\bar{\alpha}_i = g_i(-r^{2^i})$ for $i \in \{0, 1, \ldots, d-1\}$.
7. $\mathbf{V}$ sets $\alpha_d = v + \rho \cdot v^s$.
8. For $i = d-1$ to 0, $\mathbf{V}$ computes the value $\alpha_i = \texttt{derive}(u_i, \bar{\alpha}_i, \alpha_{i+1})$. (if the computation involves a zero denominator $\mathbf{V}$ aborts.)
9. $\mathbf{P}$ and $\mathbf{V}$ engage in $\texttt{SHPLONK}$ [BDFG20] and run its open procedure with inputs $\left(D, 2d, \{\mathsf{cm}_{pos}, \mathsf{cm}_{neg}, \{\mathsf{cm}_i\}_{i=1}^{d-1}\}, T = \{r^{2^i}, -r^{2^i}\}_{i=0}^{d-1}, \{S_i = \{r^{2^i}, -r^{2^i}\}\}_{i=0}^{d-1}, \{R_i\}_{i=0}^{d-1}\right)$ where $\mathsf{cm}_{pos} := \mathsf{cm} + \frac{\rho}{r}\mathsf{cm}$ and $\mathsf{cm}_{neg} := \mathsf{cm} - \frac{\rho}{r}\mathsf{cm}$, and R_is are degree-1 univariate polynomials such that $R_i(r^{2^i}) = \alpha_i$, $R_i(-r^{2^i}) = \bar{\alpha}_i$.
   Informally, $\mathbf{V}$ wishes to verify that
   - For $i \in [d-1]$, cm_i at r^{2^i} and $-r^{2^i}$ opens to α_i and $\bar{\alpha}_i$, respectively.
   - cm_{pos} opens at r to α_0, and cm_{neg} opens at $-r$ to $\bar{\alpha}_0$.
10. $\mathbf{V}$ outputs acc if the verifier of $\texttt{SHPLONK}$ in the previous step outputs acc. Else, it outputs rej.

6 Knowledge soundness proof

We define the extractor E for the knowledge soundness game from Definition 3.3.

Fix any efficient adversay $\mathcal{A}$. Suppose that given srs, $\mathcal{A}$ outputs cm, and scalars $a_0, \ldots, a_{D-1}$ such that

$$\mathsf{cm} = \sum_{i=0}^{D-1} a_i \left[x^i\right]_1$$

Then E will output the d-variate multilinear polynomial F defined by the first n coefficients: , i.e., $F := \mathfrak{ml}_d(\mathsf{F})$ where $\mathsf{F} := (a_0, \ldots, a_{n-1})$.

We wish to show that $\mathcal{A}$ wins the KS game with probability $\mathsf{negl}(\lambda)$.

For this, we look at an extended probability space defined in the following way.

- $\mathcal{A}$ and E play the KS game.
- Let $E_{\texttt{SHPLONK}}$ be the extractor of the $\texttt{SHPLONK}$ PCS. We invoke $E_{\texttt{SHPLONK}}$ on the commitments sent in Step 9 of the open protocol.

We look at a random variable containing both messages output during the KS game and by $E_{\texttt{SHPLONK}}$, and analyze events under its underlying distribution. Note in particular, that the event of $\mathcal{A}$ winning the game has the same probability in this space and in the original KS game. Thus, showing the event has probability $\mathsf{negl}(\lambda)$ in this space suffices.

Let u be randomly sampled from $\mathbb{F}^d$. $\mathcal{A}$ outputs $v \in \mathbb{F}$ and $v^s \in \mathbb{F}$.

We make the following definitions.

- Let $f(X) := \sum_{i=0}^{D-1} a_i X^i$ and recall that $f^{\leftarrow}(X) = f(X)/X$. Define $h(X) := f(X) + \rho \cdot f^{\leftarrow}(X)$. Note that $f^{\leftarrow}$ and h may be rational functions.
- Let $h_0 = h, f_0 = f$ and $f_0^s = f^{\leftarrow}$. For $i \in [d]$ define
  - $h_i := \mathtt{fold}(h_{i-1}, u_{i-1})$.
  - $f_i := \mathtt{fold}(f_{i-1}, u_{i-1})$.
  - $f_i^s := \mathtt{fold}(f_{i-1}^s, u_{i-1})$.
- Let $g_1, \ldots, g_{d-1}$ be polynomials output by $E_{\mathtt{SHPLONK}}$ on commitments $\mathsf{cm}_1, \ldots, \mathsf{cm}_{d-1}$, respectively. Let $g_d = \mathtt{fold}(g_{d-1}, u_{d-1})$. Let g_0^{pos} and g_0^{neg} denote the polynomials output by $E_{\mathtt{SHPLONK}}$ on commitments cm_{pos} and cm_{neg}, respectively.

To prove that the probability that $\mathcal{A}$ wins, i.e., **V** accepts when $v \neq F(u)$ or $v^s \neq F^{\leftarrow}(u)$, is negligible, we consider the following events and show that the probability of each of these events is negligible.

1. **Event E1:** **V** accepted and one of the following occured.
   - some evaluation of one of the g_i's for $i \in [d-1]$ or g_0^{pos} or g_0^{neg} is incorrect.
   - $g_0^{pos} \neq f + \frac{\rho}{r} f$.
   - $g_0^{neg} \neq f - \frac{\rho}{r} f$.

   The probability of the first item is $\mathsf{negl}(\lambda)$ due to the knowledge soundness of $\mathtt{SHPLONK}$.

   From linearity of $\mathtt{SHPLONK}$ commitments (identical to KZG commitments[KZG10]), $\mathsf{cm}_{pos} = \mathsf{cm} + \frac{\rho}{r}\mathsf{cm}$ is a commitment to $f + \frac{\rho}{r} f$. Thus ,when $g_0^{pos} \neq f + \frac{\rho}{r} f$ we can output a collision of KZG; and so[1] the second item has $\mathsf{negl}(\lambda)$ probability, and by similar argument so does the third.

2. **Event E2:** **V** accepted, the first event didn't occur, and for some $i \in [d-1]$, $g_i \neq h_i$.

   Let $i \in [d-1]$ denote the smallest index such that $g_i \neq h_i$.

   Case $i = 1$: $g_1 \neq h_1$.

[1]Definition 2.3 in [BDFG20] doesn't explicitly mention a binding property. One can be proven for KZG commitments under the Q-DLOG assumption: A collision allows to find a non-zero polynomial whose roots include the SRS secret x. By factoring this polynomial we can retreive x (This is similar to proofs from [FKL18].).

Let $e(X)$ be the polynomial

$$e(X) := (X\bar{u}_0 + u_0)\left(h_0(X) - 2Xg_1(X^2) + h_0(-X)\,(X\bar{u}_0 - u_0)\right).$$

We claim that given **E2**, e is not the zero polynomial. Assume otherwise. Rearranging terms we have

$$g_1(X^2) = \bar{u}_0\frac{h_0(X) + h_0(-X)}{2} + u_0\frac{h_0(X) - h_0(-X)}{2X} = h_1(X^2),$$

where the second equality is true since $h_1 = \texttt{fold}(h_0, u_0)$. But this is a contradiction since we are assuming $g_1 \neq h_1$. Thus, e is not the zero polynomial. We now show that $\mathbf{V}$ accepting implies $e(r) = 0$.

Since we're outside of the first event, we have

$$f(r) + \frac{\rho}{r}f(r) = g_0^{pos}(r) = \alpha_0 \tag{1}$$

Thus $h_0(r) = f(r) + \frac{\rho}{r}f(r) = \alpha_0$. By definition, $\alpha_0 = \texttt{derive}(u_0, \bar{\alpha}_0, \alpha_1)$. Hence,

$$h_0(r) = \texttt{derive}(u_0, \bar{\alpha}_0, \alpha_1)$$

i.e.,

$$h_0(r) = \frac{2r\alpha_1 - \bar{\alpha}_0\,(r\bar{u}_0 - u_0)}{r\bar{u}_0 + u_0}.$$

As we're outside of the first event we have $\alpha_1 = g_1(r^2)$ and $\bar{\alpha}_0 = g_0^{neg}(-r)$. One can now show $g_0(-r) = h_0(-r)$ similar to above. Substituting $\alpha_1 = g_1(r^2)$ and $\bar{\alpha}_0 = h_0(-r)$ in the previous equation, we have

$$h_0(r) = \frac{2rg_1(r^2) - h_0(-r)\,(r\bar{u}_0 - u_0)}{r\bar{u}_0 + u_0}$$

$$\implies h_0(r) - \frac{2rg_1(r^2) - h_0(-r)\,(r\bar{u}_0 - u_0)}{r\bar{u}_0 + u_0} = 0$$

$$\implies (r\bar{u}_0 + u_0)h_0(r) - 2rg_1(r^2) + h_0(-r)\,(r\bar{u}_0 - u_0) = 0$$

$$\implies e(r) = 0$$

That is, for $\mathbf{V}$ to accept, it must hold that $e(r) = 0$. By Schwartz-Zippel Lemma, this happens with probability at most $\frac{2D-1}{|\mathbb{F}|} = \mathsf{negl}(\lambda)$. This is because degree of $e = max\{2 + \deg(h_0), 2 + 2\deg(g_1)\} = max\{2 + D - 1, 2 + 2(D-1)\} = 2D$.

Case $i > 1$: $g_i \neq h_i$

Since i is the smallest index such that $g_i \neq h_i$, for all $j = 0, \ldots, i-1$, we have $g_j = h_j$.

Similarly to before define

$$e(X) := (X\bar{u}_{i-1} + u_{i-1})\left(h_{i-1}(X) - \texttt{derive}(u_{i-1}, h_{i-1}(-X), g_i(X^2))\right)$$
$$= (X\bar{u}_{i-1} + u_{i-1})\left(h_{i-1}(X) - 2Xg_i(X^2) + h_{i-1}(-X)\,(X\bar{u}_{i-1} - u_{i-1})\right)$$

As in the case $i = 1$, one can show e is not the zero polynomial when $g_i \neq h_i$.

When **V** accepts and we're outside of **E1**

$$g_{i-1}(r^{2^{i-1}}) = \alpha_{i-1}$$

Since $g_{i-1} = h_{i-1}$, we have $g_{i-1}(r^{2^{i-1}}) = h_{i-1}(r^{2^{i-1}})$. Further, recall that $\alpha_{i-1} = \texttt{derive}(u_{i-1}, \bar{\alpha}_{i-1}, \alpha_i)$. Hence, substituting these in the previous equation it must hold that

$$h_{i-1}(r^{2^{i-1}}) = \texttt{derive}(u_{i-1}, \bar{\alpha}_{i-1}, \alpha_i)$$

i.e.,

$$h_{i-1}(r^{2^{i-1}}) = \frac{2r^{2^{i-1}}\alpha_i - \bar{\alpha}_{i-1}\left(r^{2^{i-1}}\bar{u}_{i-1} - u_{i-1}\right)}{r^{2^{i-1}}\bar{u}_{i-1} + u_{i-1}}$$

We also have in our case $\alpha_i = g_i(r^{2^i})$ and $\bar{\alpha}_{i-1} = g_{i-1}(-r^{2^{i-1}}) = h_{i-1}(-r^{2^{i-1}})$. Thus,

$$h_{i-1}(r^{2^{i-1}}) = \frac{2r^{2^{i-1}}g_i(r^{2^i}) - h_{i-1}(-r^{2^{i-1}})\left(r^{2^{i-1}}(1 - u_{i-1}) - u_{i-1}\right)}{r^{2^{i-1}}\bar{u}_{i-1} + u_{i-1}}$$

$$\implies \left(r^{2^{i-1}}\bar{u}_{i-1} + u_{i-1}\right)h_{i-1}(r^{2^{i-1}}) - 2r^{2^{i-1}}g_i(r^{2^i})$$
$$+ h_{i-1}(-r^{2^{i-1}})\left(r^{2^{i-1}}\bar{u}_{i-1} - u_{i-1}\right) = 0$$

$$\implies e(r^{2^{i-1}}) = 0$$

That is, for **V** to accept, it must hold that $e(r^{2^{i-1}}) = 0$. Observe here that degree of $e = max\{2+\deg(h_{i-1}), 2+2\deg(g_i)\} = max\{2+(D-1)/2^{i-1}, 2+2(D-1)\} = 2D$. Thus, degree of $e(X)$ is at most $2D$. Plugging $X = r^{2^{i-1}}$ multiplies the degree by at most 2^{i-1}. Thus, by Schwartz-Zippel Lemma, probability that $e(r^{2^{i-1}}) = 0$ is at most $\frac{2^{i-1}(2D)}{|\mathbb{F}|} \leq \frac{D^2}{|\mathbb{F}|} = \mathsf{negl}(\lambda)$.

3. **Event E3: V** accepted, first and second events didn't occur, and $g_d(X) \not\equiv v + \rho\cdot v^s$.

   As before, since $E_{\texttt{SHPLONK}}$ returns polynomials with correct evaluations outside **E1**, we have $\alpha_{d-1} = g_{d-1}(r^{2^{d-1}})$ and $\bar{\alpha}_{d-1} = g_{d-1}(-r^{2^{d-1}})$. Recall that

$$\alpha_{d-1} = \texttt{derive}(u_{d-1}, \bar{\alpha}_{d-1}, \alpha_d) \tag{2}$$

Since $\alpha_{d-1} = g_{d-1}(r^{2^{d-1}})$, $\bar{\alpha}_{d-1} = g_{d-1}(-r^{2^{d-1}})$ and $\alpha_d := v + \rho \cdot v^s$, we have

$$g_{d-1}(r^{2^{d-1}}) = \texttt{derive}(u_{d-1}, g_{d-1}(-r^{2^{d-1}}), v + \rho \cdot v^s)$$

$$\iff g_{d-1}(r^{2^{d-1}}) = \frac{2r^{2^{d-1}}(v+\rho\cdot v^s) - g_{d-1}(-r^{2^{d-1}})\left(r^{2^{d-1}}(1-u_{d-1}) - u_{d-1}\right)}{r^{2^{d-1}}(1-u_{d-1}) + u_{d-1}}$$

$$\iff v+\rho\cdot v^s = (1-u_{d-1})\frac{g_{d-1}(r^{2^{d-1}}) + g_{d-1}(-r^{2^{d-1}})}{2} + u_{d-1}\frac{g_{d-1}(r^{2^{d-1}}) - g_{d-1}(-r^{2^{d-1}})}{2r^{2^{d-1}}}$$

$$\iff v + \rho \cdot v^s = g_d(r^{2^d})$$

where the last implication uses $g_d = \texttt{fold}(g_{d-1}, u_{d-1})$, and hence it follows that $g_d(X^2) = \bar{u}_{d-1}\frac{g_{d-1}(X)+g_{d-1}(-X)}{2} + u_{d-1}\frac{g_{d-1}(X)-g_{d-1}(-X)}{2X}$. Thus, $\alpha_{d-1} = g_{d-1}(r^{2^{d-1}})$ if and only if $v + \rho \cdot v^s = g_d(r^{2^d})$.

Thus, in our current case, using $g_d(X) \not\equiv v + \rho \cdot v^s$

$$\begin{aligned}\Pr[\mathbf{V} \text{ accepts}] &\leq \Pr\left[g_d(r^{2^d}) = v + \rho \cdot v^s\right] \\ &\leq \frac{2^d(D-1)/2^d}{|\mathbb{F}|} = \frac{(D-1)}{|\mathbb{F}|} = \mathsf{negl}(\lambda)\end{aligned}$$

For the last inequality, observe that the degree of g_d can be at most $(D-1)/2^d$. This is because outside the second event we know that all the foldings $g_i = h_i$ for $i \in [d-1]$ are computed correctly and $g_d = \texttt{fold}(g_{d-1}, u_{d-1})$. The adversary may have started with polynomial g_0 of degree at most $D-1$, and for each $i \in [d]$, we have $2\deg(g_i) \leq \deg(g_{i-1})$.

Therefore, $\deg(g_d)$ is at most $(D-1)/2^d$. Since plugging $X = r^{2^d}$ multiplies the degree by at most 2^d, by Schwartz-Zippel Lemma, probability that $g_d(r^{2^d}) = v + \rho \cdot v^s$ is at most $\frac{2^d(D-1)/2^d}{|\mathbb{F}|} = \frac{(D-1)}{|\mathbb{F}|} = \mathsf{negl}(\lambda)$.

4. **Event E4:** $\mathbf{V}$ accepted, the first, second and third events didn't occur, and $f_d \not\equiv v$ or $f_d^s \not\equiv v^s$.

   When $\mathbf{V}$ accepts, outside the first three events we have, $g_i = h_i$ for $i \in [d-1]$ and $g_d \equiv v+\rho\cdot v^s$. From the definition of $g_d = \texttt{fold}(g_{d-1}, u_{d-1}), h_d = \texttt{fold}(h_{d-1}, u_{d-1})$, we have

$$g_d = \texttt{fold}(g_{d-1}, u_{d-1}) = \texttt{fold}(h_{d-1}, u_{d-1}) = h_d$$

   Since $g_d \equiv v + \rho \cdot v^s$, we have $h_d = g_d \equiv v + \rho \cdot v^s$. Further, since $h = f + \rho \cdot f^{\leftarrow}$, by linearity of folding, we have $h_d = f_d + \rho \cdot f_d^s$. Hence,

$$f_d + \rho \cdot f_d^s \equiv v + \rho \cdot v^s. \tag{3}$$

   Or equivalently

$$(f_d - v) + \rho(f_d^s - v^s) \equiv 0. \tag{4}$$

   If $f_d \not\equiv v$ or $f_d^s \not\equiv v^s$ this can hold for at most one $\rho \in \mathbb{F}$, or equivalently, with probability at most $1/|\mathbb{F}|$ over ρ.

5. **Event E5:** **V** accepts, $E1 - E4$ didn't occur, and $a_n \neq 0$.

   Denote $f_d(X) = \sum_{i=0}^{D-1} b_i X^i$. Note that we can think of the coefficients b_i as polynomials in u. The term a_n shows up in the coefficient of the linear term b_1. More precisely,

$$b_1(u) = \sum_{j=0}^{n-1} a_{n+j} L_j(u)$$

   This is a non-zero polynomial in u when $a_n \neq 0$ because of the first term in the sum and the linear independence of the L_j. By the Schwartz-Zippel lemma, the probability that $b_1(u) = 0$ is at most $d/|\mathbb{F}|$ over u. But we know that outside of **E4** we have $b_1 = 0$.

Now we prove two claims that will enable us to conclude that outside the above five events $\mathcal{A}$ loses the KS game. We first define some notation we will use.

Notation for claims: Let $G(X_0, \ldots, X_{k-1})$ be the $k = \log(D)$-variate multilinear defined by *all* of f's D coefficients. Namely, $G := \mathfrak{ml}_k(f)$. Accordingly, define $G^{\leftarrow}(X_0, \ldots, X_{k-1}) := \mathfrak{ml}_k(h)$ where $h := (a_1, \ldots, a_{D-1}, 0)$. We will need to refer to the coefficients of f_d and f_d^s. For this purpose, let $D' := 2^{k-d} - 1$ and denote

$$f_d(X) = \sum_{i=0}^{D'} b_i X^i, f_d^s(X) = \sum_{i=-1}^{D'} c_i X^i.$$

Claim 6.1. $F(u) = f_d(0)$.

Proof. From Claim 3.1 we have that $F(u) = G(u, 0^{k-d})$. Now, we show that $G(u, 0^{k-d}) = b_0$: Recall we defined G_u earlier (in Section 4) as

$$G_u(X_d, \ldots, X_{k-1}) = G(u_0, \ldots, u_{d-1}, X_d, \ldots, X_{k-1}).$$

As explained in Section 4, we have that $G_u = \mathfrak{ml}_{k-d}(f_d)$.

That is,

$$G_u(X_d, \ldots, X_{k-1}) = \sum_{i=0}^{D'} b_i L'_i(X_d, \ldots, X_{k-1}),$$

where L'_i is the $k-d$ dimensional i'th Lagrange polynomial. Explicitly, $L'_i(X_d, \ldots, X_{k-1}) := \Pi_{j=d}^{k-1} \left(i_j X_j + \bar{i}_j \bar{X}_j \right)$, for $i_d, \ldots, i_{k-1}$ that are the binary representation of $0 \leq i < 2^{k-d}$.

In particular,

$$G(u, 0^{k-d}) = G_u(0, \ldots, 0) = b_0 = f_d(0).$$

In summary, we have

$$F(u) = G(u, 0^{k-d}) = f_d(0).$$

□

Claim 6.2. $F^{\leftarrow}(u) = f_d^s(0) - a_n \prod_{0 \leq i < d} u_i$.

Proof. From Claim 3.2 we have that $F^{\leftarrow}(u) = G^{\leftarrow}(u, 0^{k-d}) - a_n \prod_{0 \le i < d} u_i$. Now, we show that $G^{\leftarrow}(u, 0^{k-d}) = f_d^s(0)$.

Let
$$H^s(X_d, \ldots, X_{k-1}) := G^{\leftarrow}(u_0, \ldots, u_{d-1}, X_d, \ldots, X_{k-1}).$$

As explained in Section 4, we have $H^s = \mathfrak{ml}_{k-d}(f_d^s)$.

That is,
$$H^s(X_d, \ldots, X_{k-1}) = \sum_{i=0}^{D'} c_i L'_i(X_d, \ldots, X_{k-1})$$
where L'_i is the $k - d$ dimensional i'th Lagrange polynomial, $L'_i(X_d, \ldots, X_{k-1}) := \Pi_{j=d}^{k-1} \left(i_j X_j + \bar{i}_j \bar{X}_j \right)$, for $i_d, \ldots, i_{k-1}$ that are the binary representation of $0 \le i < 2^{k-d}$.

In particular,
$$G^{\leftarrow}(u, 0^{k-d}) = H^s(0, \ldots, 0) = c_0 = f_d^s(0).$$

Thus, we have $F^{\leftarrow}(u) = G^{\leftarrow}(u, 0^{k-d}) - a_n \prod_{0 \le i < d} u_i = f_d^s(0) - a_n \prod_{0 \le i < d} u_i$. □

Outside E1-E5, if **V** accepts, then from Claim 6.1, Claim 6.2, we have that:
$$F(u) = v, \quad F^{\leftarrow}(u) = v^s. \tag{5}$$

Thus, the knowledge soundness error comes from

- E1 (SHPLONK knowledge soundness error): happens with probability $\mathsf{negl}(\lambda)$.
- E2 (incorrect folding): happens with probability at most $\frac{D^2}{|\mathbb{F}|}$.
- E3 (g_d not being a constant): happens with probability at most $\frac{D-1}{|\mathbb{F}|}$.
- E4 ($f_d(X) \not\equiv v$ or $f_d^s \not\equiv v^s$): happens with probability at most $\frac{1}{|\mathbb{F}|}$.
- E5 ($a_n \ne 0$ but **V** accepts): happens with probability at most $\frac{d}{|\mathbb{F}|}$.

Putting it together,
$$\begin{aligned}
\Pr[\mathbf{V} \text{ accepts } &\wedge (v \ne F(u) \vee v^s \ne F^{\leftarrow}(u))] \\
&\le \mathsf{negl}(\lambda) + \frac{D^2}{|\mathbb{F}|} + \frac{(D-1)}{|\mathbb{F}|} + \frac{1}{|\mathbb{F}|} + \frac{d}{|\mathbb{F}|} \\
&= \mathsf{negl}(\lambda)
\end{aligned}$$

The case of multiple polynomials: One may wish to evaluate several polynomials and their shifts at the same point. To account for such a batched scenario, we follow the standard approach of taking a random linear combination of the polynomials and their commitments, and use that as input to the protocol. Elaborately, to handle p plain evaluations $\{f^{(j)}(u)\}_{j=1}^{p}$ and q shifted evaluations $\{f^{(j)\leftarrow}(u)\}_{j=1}^{q}$ in one shot, the prover aggregates with a single challenge $\rho \in \mathbb{F}$, and defines

$$F_\rho(X) = \sum_{j=1}^{p} \rho^j f^{(j)}(X), \quad G_\rho(X) = \sum_{j=1}^{q} \rho^{p+j} f^{(j)}(X), \quad g_0(X) = F_\rho(X) + \frac{G_\rho(X)}{X}.$$

Prover runs the folding steps to compute $g_i := \mathtt{fold}(g_{i-1}, u_{i-1})$ for $i \in [d]$, and sends the commitments $\mathsf{cm}_i = \mathsf{com}(g_i, \mathsf{srs})$, for $i = 1, \ldots, d-1$. Upon receiving the challenge r, it also sends $\bar{\alpha}_i = g_i(-r^{2^i})$, for $i = 1, \ldots, d-1$. The verifier sets the batched target $\alpha_d = \sum_{j=1}^{p} \rho^j v_j + \sum_{j=1}^{q} \rho^{p+j} w_j$ from the claimed values $\{v_j\}, \{w_j\}$, and derives α_i backwards. Finally, it computes $C_U := \sum_{j=1}^{p} \rho^j \mathsf{cm}_j^U$ and $C_S := \sum_{j=1}^{q} \rho^{p+j} \mathsf{cm}_j^S$, where cm_j^U is a commitment to $f^{(j)}$ from the $f^{(j)}$'s that remain unshifted, and cm_j^S is a commitment to $f^{(j)}$ from the ones to be shifted. It verifies that

- $C_0^{pos} := C_U + r^{-1} C_S$ opens at r to α_0;
- $C_0^{neg} := C_U - r^{-1} C_S$ opens at $-r$ to $\bar{\alpha}_0$;
- for each $i = 1, \ldots, d-1$, the commitment cm_i opens at r^{2^i} to α_i and at $-r^{2^i}$ to $\bar{\alpha}_i$.

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References

[azt] https://github.com/aztecprotocol/aztec-packages/.

[BCHO22] J. Bootle, A. Chiesa, Y. Hu, and M. Orrù. Gemini: Elastic snarks for diverse environments. *IACR Cryptol. ePrint Arch.*, page 420, 2022.

[BDFG20] D. Boneh, J. Drake, B. Fisch, and A. Gabizon. Efficient polynomial commitment schemes for multiple points and polynomials. *IACR Cryptol. ePrint Arch.*, page 081, 2020.

[CBBZ23] B. Chen, B. Bünz, D. Boneh, and Z. Zhang. Hyperplonk: Plonk with linear-time prover and high-degree custom gates. In *Advances in Cryptology - EUROCRYPT 2023 - 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques, Lyon, France, April 23-27, 2023, Proceedings, Part II*, pages 499–530, 2023.

[FKL18] G. Fuchsbauer, E. Kiltz, and J. Loss. The algebraic group model and its applications. In *Advances in Cryptology - CRYPTO 2018 - 38th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 19-23, 2018, Proceedings, Part II*, pages 33–62, 2018.

[GL25] L. Garms and M. Livesey. Attacking soundness for an optimization of the gemini polynomial commitment scheme. Cryptology ePrint Archive, Paper 2025/565, 2025.

[ham] https://hackmd.io/@adrian-aztec/bjxoyecqj.

[KZG10] A. Kate, G. M. Zaverucha, and I. Goldberg. Constant-size commitments to polynomials and their applications. In *Advances in Cryptology - ASIACRYPT 2010 - 16th International Conference on the Theory and Application of Cryptology and Information Security, Singapore, December 5-9, 2010. Proceedings*, pages 177–194, 2010.